

WASP-36b

R-1360

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_190124 · created 2026-07-31T23:34:01+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458507.7434 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.015
Transit depth (Rp/Rs)^2	1.92 +/- 0.43 [%]
Orbital Inclination (inc)	84.71 +/- 1.28
Airmass coefficient 1 (a1)	0.987 +/- 0.016
Airmass coefficient 2 (a2)	0.0062 +/- 0.0051
Scatter in the residuals of the lightcurve fit is	1.6 %
Best Comparison Star	#2 - [451, 58]
Optimal Method	PSF photometry
Transit Duration (day)	0.0838 +/- 0.0062

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.139 ± 0.015 vs 0.1368 (deviation 0.13σ)
Duration fit vs published	0.0838 d vs 0.0758 d (deviation 1.13σ)
Mid-time O–C	-0.2 min
Depth SNR	8.2
Residual scatter	1.6 %

Light curve

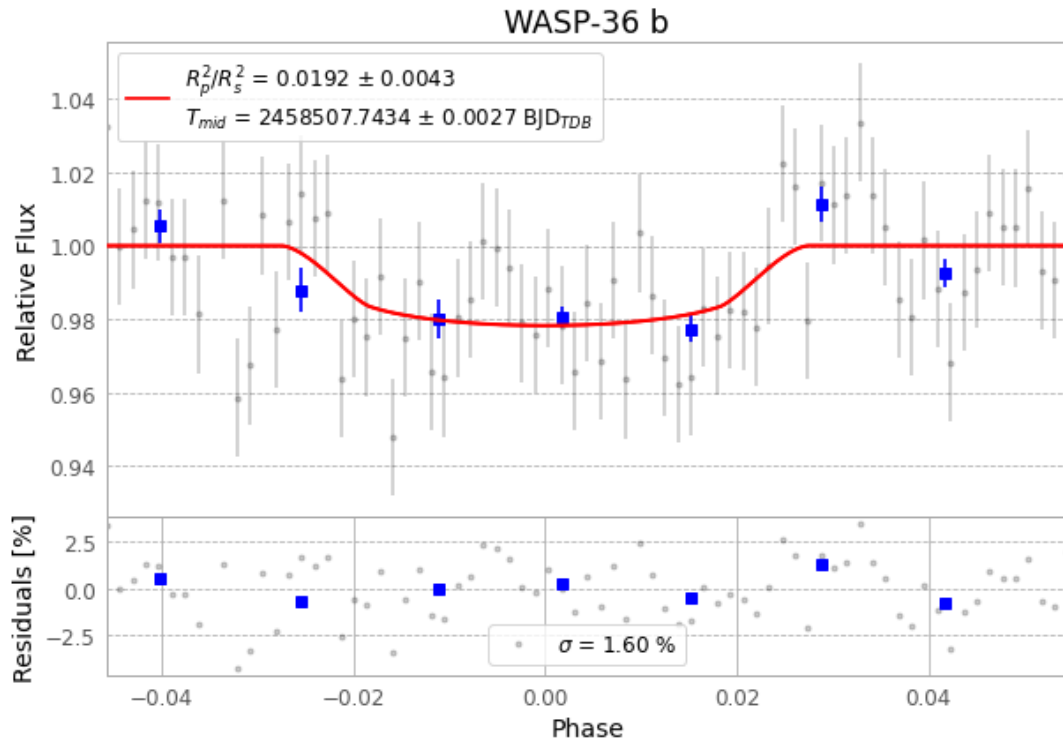


Figure 1 — FinalLightCurve_WASP-36 b_2019-01-23.png (SHA-256 6fb301c01a67694a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [383, 90], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 42a924f5ef56b65e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), WASP-36190124040012.FITS → WASP-36190124074212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-190124.log (d5c1104923f8...), FinalLightCurve_WASP-36 b_2019-01-23.png (6fb301c01a67...), FinalLightCurve_WASP-36 b_2019-01-23.csv (263ef0cce677...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 23:34Z.